

WEEK 1 – LECTURE 1

Artificial Intelligence Foundations

Unit I: What is Artificial Intelligence?

 CSE276

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Lovely Professional University



Why Are We Studying AI?



Self-Driving Cars



Medical Diagnosis AI

What do all these have in common?

Icebreaker Activity

What AI apps did you use today?



Instagram



YouTube



Spotify



Maps



Alexa

What is Artificial Intelligence?

 Human Intelligence



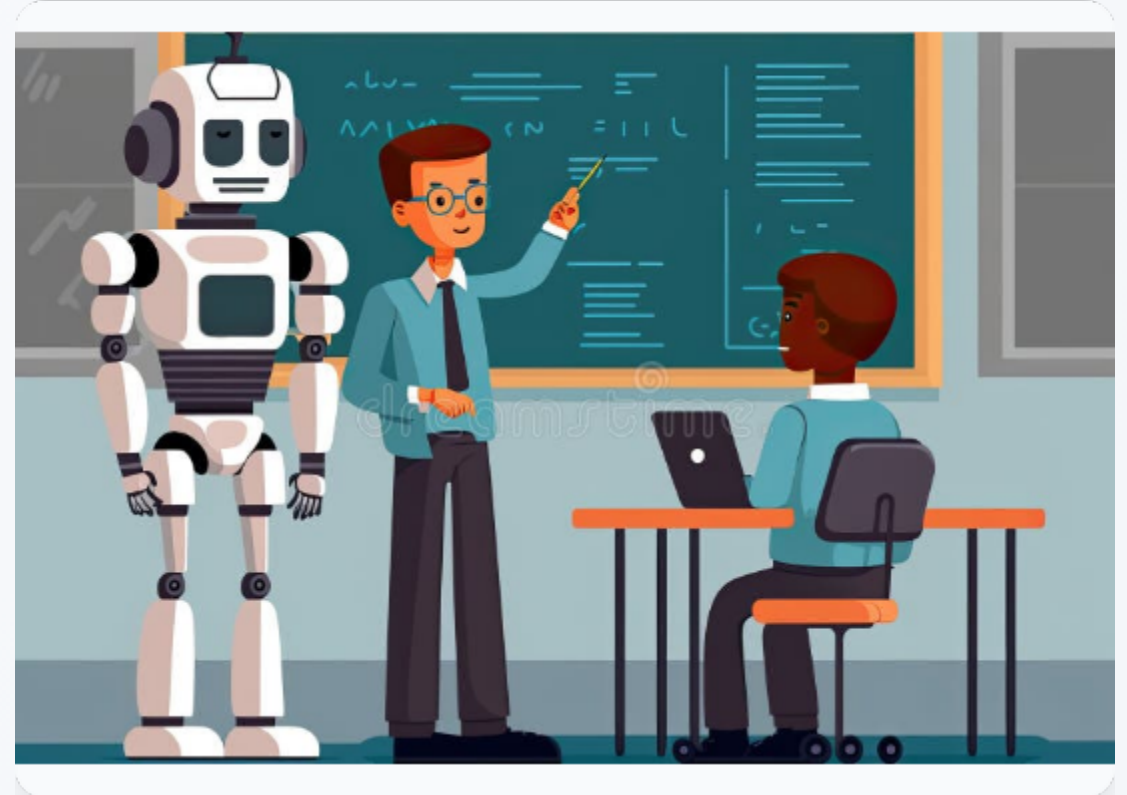
 Computer Learns



 Reasons & Decides



 Acts



Analogy: Teaching a child

Characteristics of AI



Learning



Reasoning



Planning



Perception



Problem Solving



Decision Making

Brief History of AI



Interactive Discussion

"Why did AI fail during the AI Winter?"



Types of AI

 ANI (Narrow AI)



 AGI (General AI)



 ASI (Super AI)

Narrow AI (ANI)

Excellent at ONE task.



Designed to perform specific, constrained tasks very well.

Everyday Examples

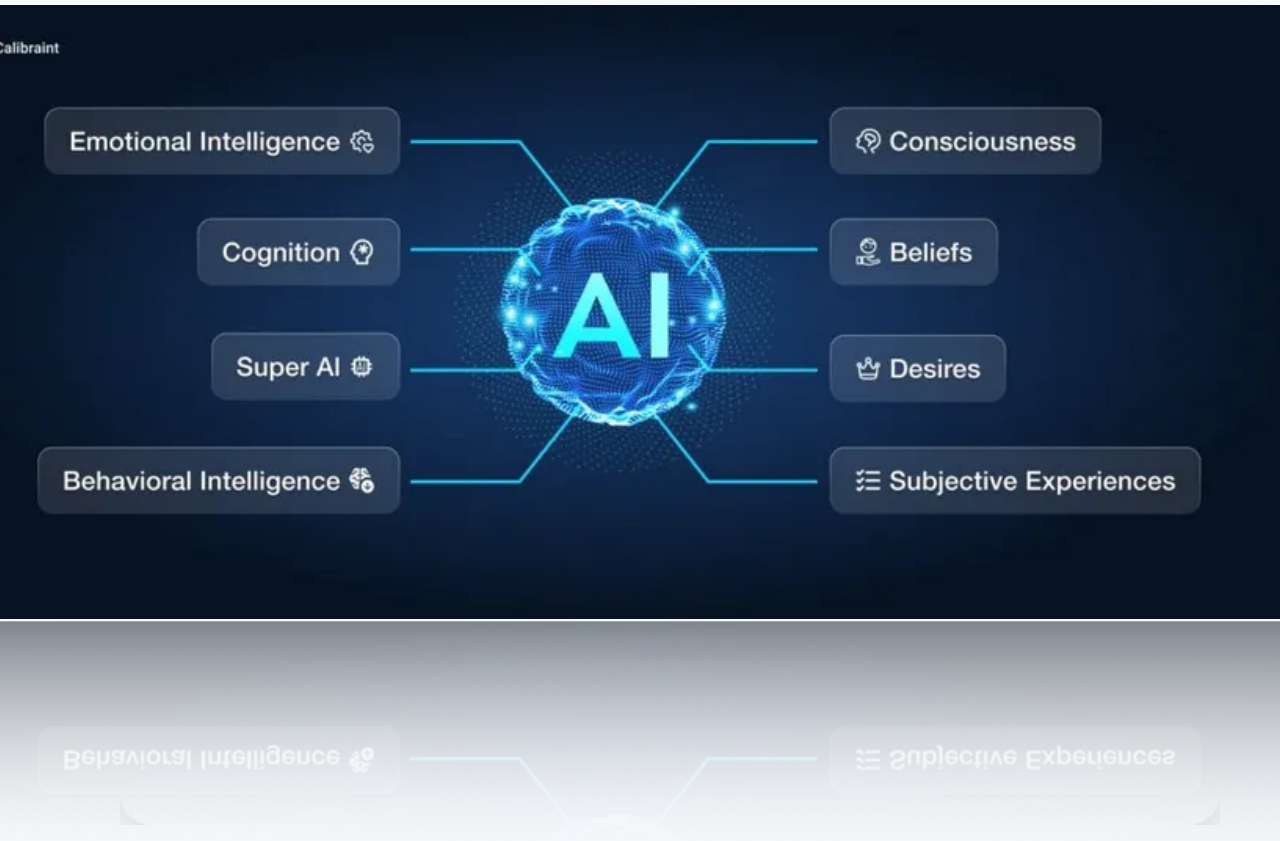
-  Virtual Assistants
-  Chess Engines
-  Recommendation Systems
-  Email Spam Filters
-  Self-Driving Algorithms

Artificial General Intelligence (AGI)

- ✓ Learns like a human.
- ✓ Reasons across multiple domains.
- ✓ Solves unfamiliar problems.
- ✓ Completely hypothetical today.
- ✗ **Not achieved yet.**



Artificial Super Intelligence (ASI)



-  Surpasses human intelligence.
-  Infinite creative capability.
-  Perfect problem solving.
-  Highly hypothetical concept.
-  Raises major ethical questions.

Interactive Poll

Which category does ChatGPT belong to?

A) ANI

B) AGI

C) ASI

Interactive Poll

Which category does ChatGPT belong to?

A) ANI



B) AGI

C) ASI

AI vs Machine Learning

Artificial Intelligence

Machine Learning



Real-World Analogy

"AI is the **entire hospital**.
Machine Learning is **one specialized doctor**."

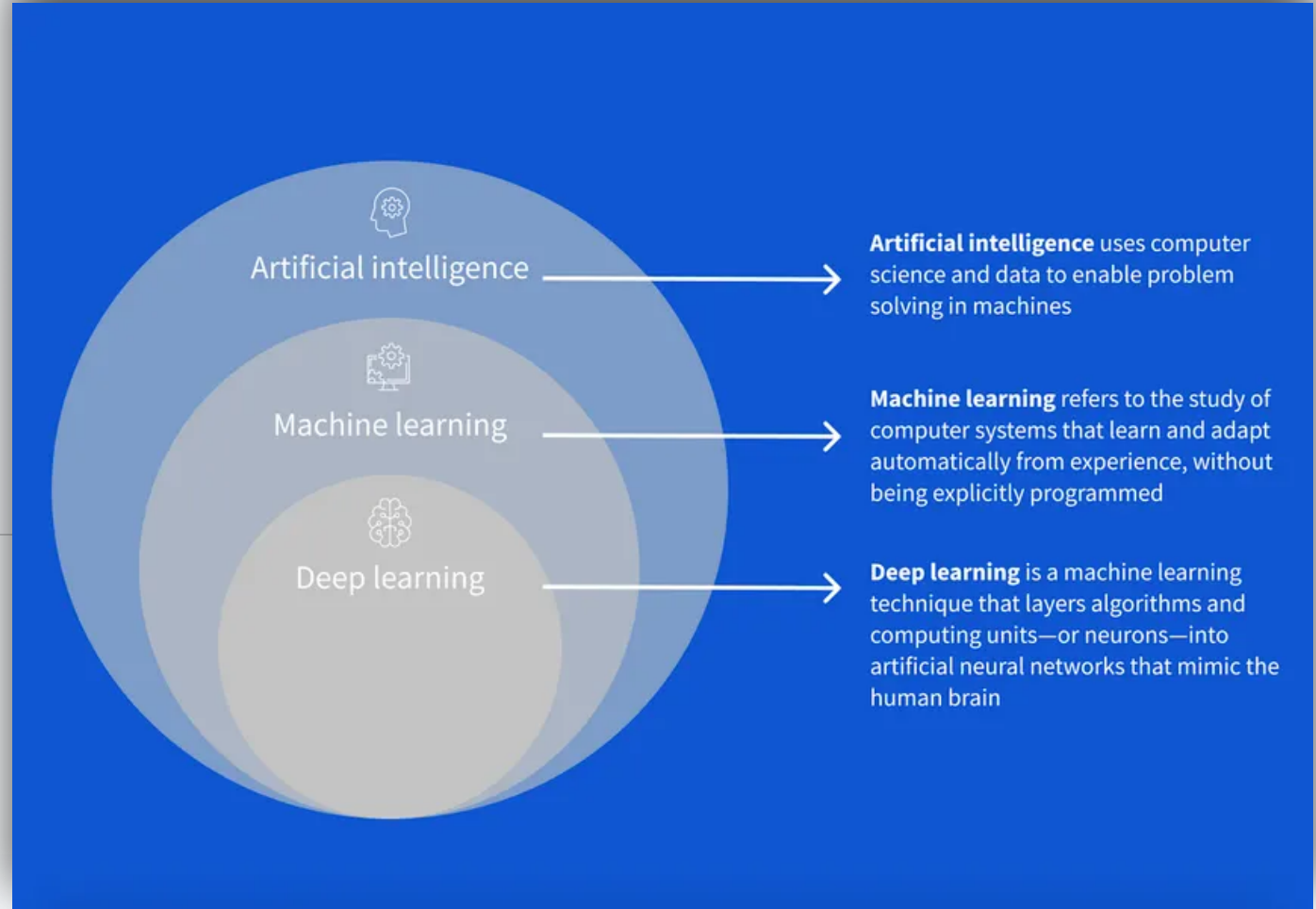
Machine Learning vs Deep Learning

 **Vehicle = AI**

 **Car = ML**

 **Electric Car = DL**

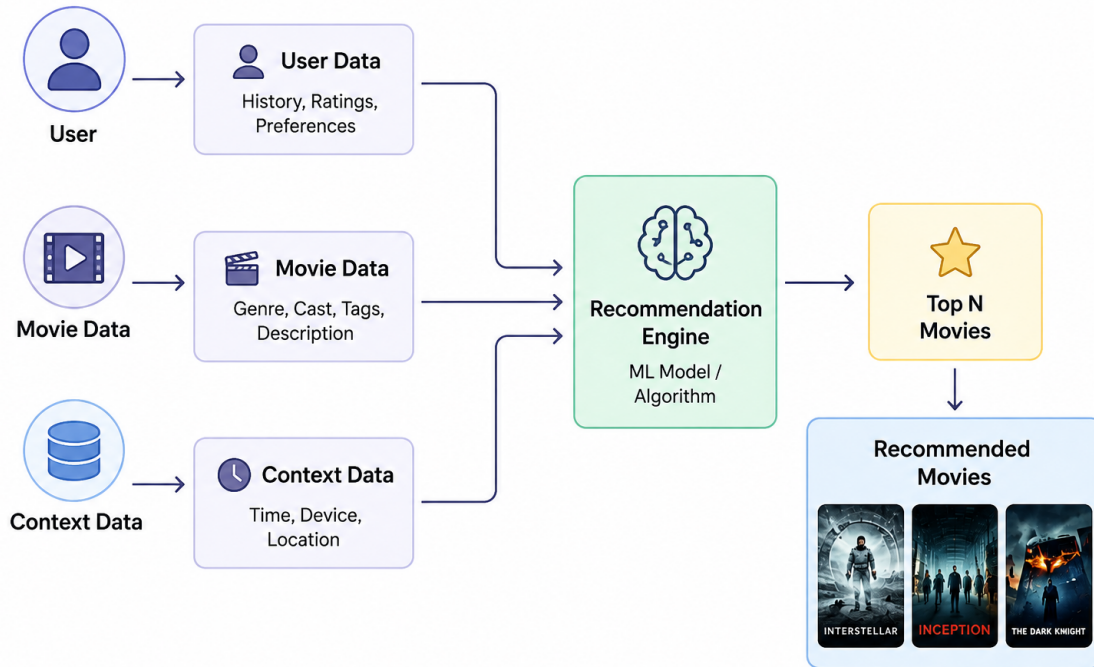
Deep Learning is a subset of ML using multi-layered artificial neural networks.



AI vs ML vs DL vs Data Science

Field	Primary Goal	Output	Example
Data Science	Extract insights from data	Reports & Dashboards	Market Analysis
AI	Mimic human intelligence	Automated Actions	Game AI Bots
Machine Learning	Learn without programming	Predictions	Spam Filter
Deep Learning	Handle complex, raw data	Advanced Patterns	Face Recognition

Case Study: Netflix Recommendations



 **User Watches Movies**



 **System Collects Data**

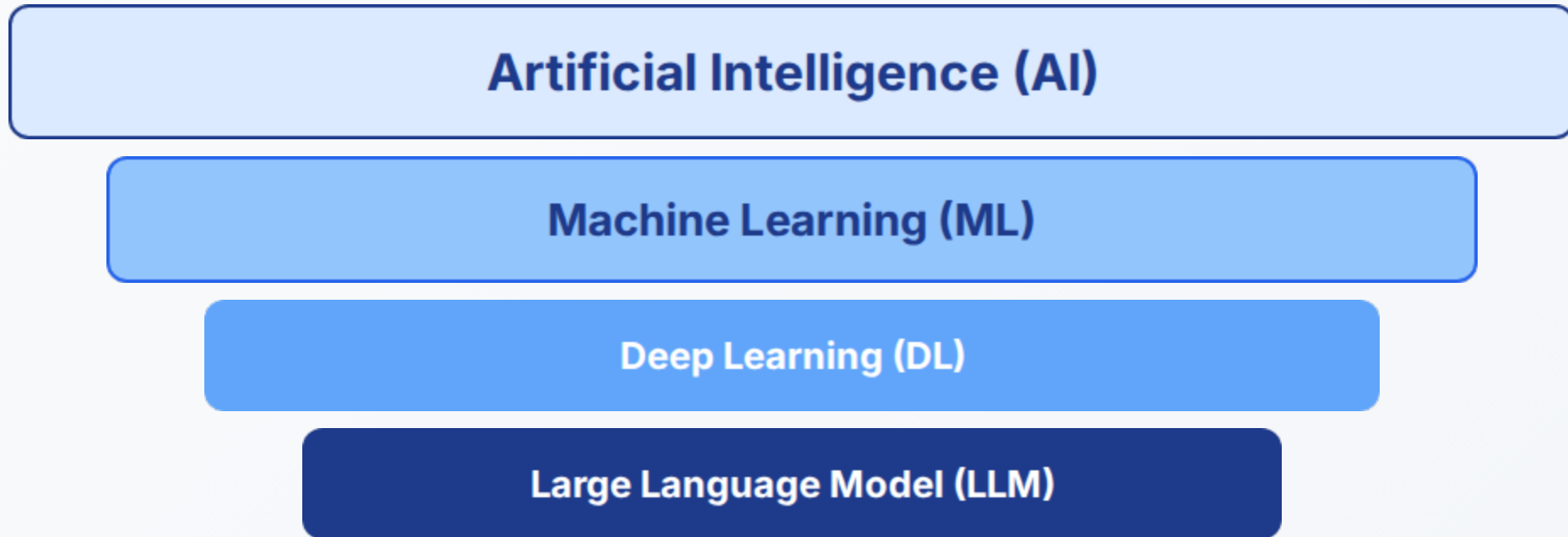


 **Machine Learning Learns**



 **Delivers Recommendation**

Where Does ChatGPT Fit?



ChatGPT is a highly specialized application of Deep Learning.



Board Work

"Let's build the AI family tree together."



Key Takeaways

- ✓ AI is much broader than ML.
- ✓ ML is much broader than DL.
- ✓ Data Science strongly overlaps AI.
- ✓ Most AI today is Narrow AI.
- ✓ AI learns, reasons, and acts.

Quick Quiz

1. Which of the following is an example of AGI?

- ☐ A) Google Maps Navigation
- ☐ B) Netflix Recommendations
- ☐ C) None of the above
- ☐ D) Face Unlock

Quick Quiz

1. Which of the following is an example of AGI?

- ☐ A) Google Maps Navigation
- ☐ B) Netflix Recommendations
- ☒ C) **None of the above**
- ☐ D) Face Unlock

Coming up...

-  The complete AI Lifecycle.
-  AI Problem Solving methods.
-  Advanced Applications of AI.



Thank You!

Any Questions?

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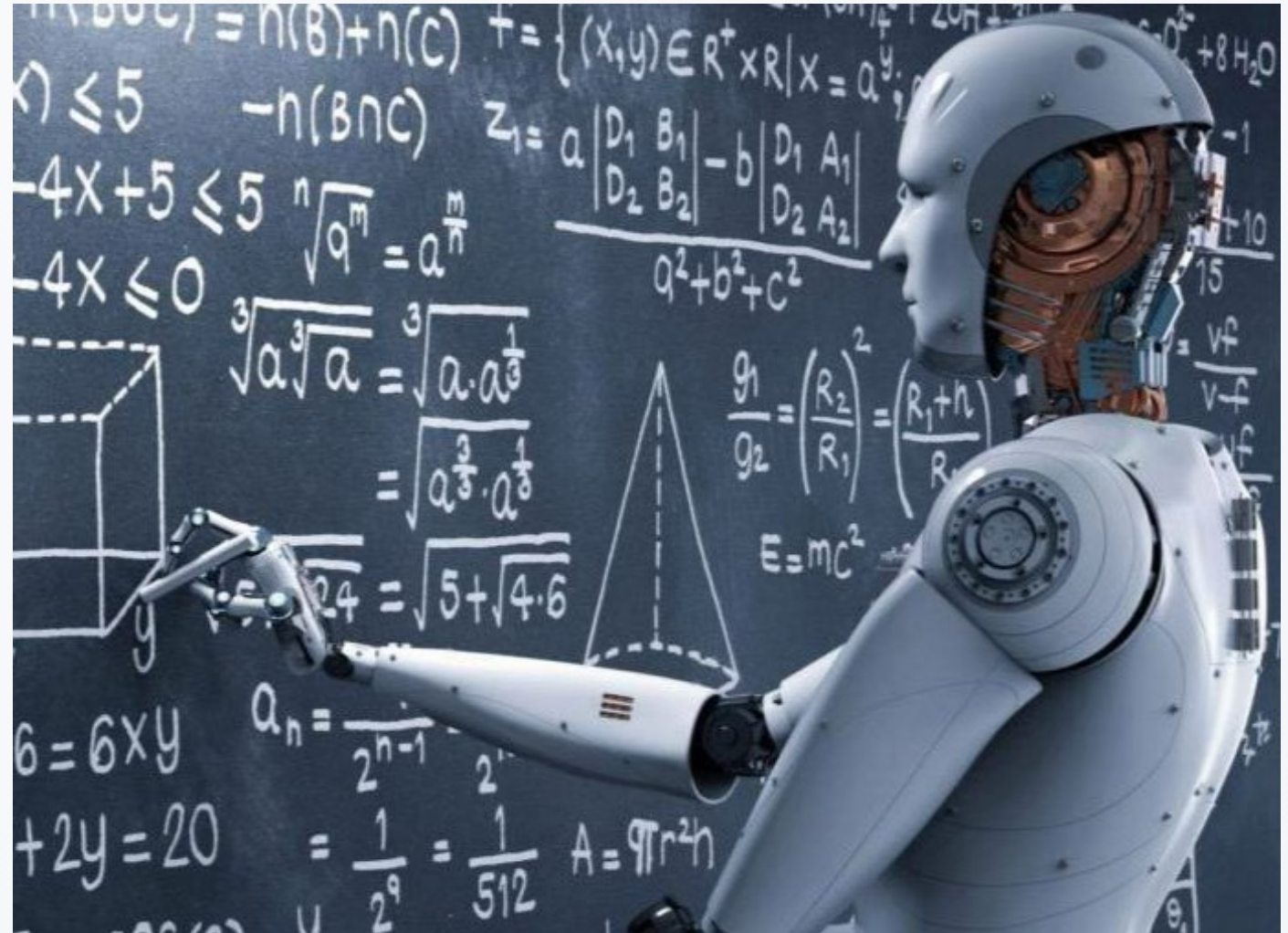


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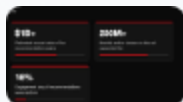
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